Hardik Pajiala

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Summary

Computer Engineering undergraduate (2028) seeking **Software and Data Engineering Intern** roles; won first place at **BIOS 2026 hackathon** and built **2M+ record Delta Lake** platform on PySpark, **serverless AWS** pipeline and live **PyTorch** sign-language recognizer

Education

Thapar Institute of Engineering & Technology CGPA: 8.50 / 10

Bachelor of Engineering in Computer Engineering *2024 – Expected June 2028*

- **Coursework:** Data Structures & Algorithms, DBMS, OOP, Software Engineering, Statistics, Machine Learning, Deep Learning

Experience & Achievements

SafeTrail – First Place, BIOS 2026 Hackathon Thapar Institute, Patiala

Top contributor in 4-person team | FastAPI, React, TypeScript, MapLibre, LLM APIs *Oct 2026 – Nov 2026*

- Developed trail-safety **PWA** for trekkers with geofenced danger alerts, **offline-queued SOS** and live responder map
- Isolated AI from emergency path: geofenced alerts, SOS and triage run as deterministic code with **Ed25519**-signed passes and event log, while LLM features get per-job budgets, standard-text fallbacks and prompt-injection flags; **86 backend tests**
- Benchmarked 4 builds on 100-trip simulation, cutting LLM usage to **82% fewer tokens per trip** than naive build

Smriti – Smart India Hackathon (SIH26003) MDoNER problem statement

Team project | React, TypeScript, Web Audio API, Flutter, SQLite *Sep 2026*

- Built memory and care companion for elders with dementia in North East India: read-aloud elder screen, Bihu rhythm game, **caregiver dashboard** with 2 trend charts and geofence map, plus offline **Flutter** tablet build on SQLite storage

Projects

Parchi – Serverless Document Pipeline | *AWS Lambda, SQS, DynamoDB, Textract* Nov 2026

- Architected **event-driven serverless pipeline** for AWS (team of 3): S3 upload events to **SQS**, **Lambda** + Textract bill extraction, DynamoDB and SNS, with **idempotent** retries, dead-letter queue and SAM infrastructure as code; **118 tests**

Ishara – Sign Language Recognition | *PyTorch, MediaPipe, BiLSTM* | Live demo Oct 2026

- Created live webcam recognizer for 61 Indian Sign Language words using **BiLSTM with attention** on MediaPipe keypoints, with speech output; replaced saturated benchmark with harder protocols reaching **95.6%** top-1 from 5 examples per word

Cricket Win-Probability Lakehouse | *PySpark, Delta Lake, Spark MLlib, Airflow* | Live demo Aug 2026 – Sep 2026

- Engineered **medallion lakehouse** on Delta Lake turning 8,852 real T20 matches of nested JSON into **2.03M ball-by-ball rows**
- Implemented exactly-once ingestion with **Spark Structured Streaming** and idempotent **Delta MERGE** upserts (0 lost records, 0 duplicate keys), guarded by 19 declarative **data-quality checks** that quarantine bad rows and halt downstream tasks
- Trained **Spark MLlib** win-probability model (**0.844 AUC**, 1.3 pp calibration error on unseen 2025–26 matches), served live via Spark `applyInPandasWithState`, orchestrated by **Airflow** DAGs (3-minute full rebuild) with GitHub Actions CI

FoodBridge – Backend & Matching Engine | *Node.js, TypeScript, PostgreSQL/PostGIS, FastAPI* Jul 2026 – Sep 2026

- Owned the backend of a 3-person food-redistribution platform: **26 REST endpoints** over a 10-table PostgreSQL schema, with JWT auth, role-based guards and state machines enforcing every donation and pickup transition.
- Designed a **FastAPI** matching service scoring donor–recipient pairs on urgency, proximity and capacity, assigning them optimally with the **Hungarian algorithm** over **PostGIS** spatial distances; covered API and matcher with Vitest and pytest tests.

Technical Skills

Languages: Python, SQL, TypeScript/JavaScript, Java, C/C++, R
Big Data: Apache Spark (PySpark, Structured Streaming, Spark SQL), Delta Lake, Kafka, Parquet, Airflow
ML & AI: PyTorch, TensorFlow/Keras, Spark MLlib, MediaPipe, BiLSTMs, LLM agents & guardrails
Databases & Backend: PostgreSQL, PostGIS, MySQL, Node.js, Express, FastAPI, REST APIs, React, PWAs
Cloud & Tools: AWS (Lambda, SQS, DynamoDB), Pandas, NumPy, Streamlit, pytest, Vitest, Git, GitHub Actions (CI/CD), Linux
Concepts: Medallion lakehouse architecture, ETL/ELT, data modelling, data quality, idempotent upserts, orchestration

Positions of Responsibility

Batch Group Leader, Thapar University | *faculty liaison for 300-student batch* 2025 – Present
Volunteer, Saturnalia (annual college fest) | *coordinated stage-event logistics with 25 volunteers* 2025